Household Heterogeneity and the Transmission of Foreign Shocks

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Abstract

We study the role of heterogeneity in the transmission of foreign shocks. We build a Heterogeneous-Agent New-Keynesian Small Open Model Economy (HANKSOME) that experiences a current account reversal. Households' portfolio composition and the extent of foreign currency borrowing are key determinants of the magnitude of the contraction in consumption associated with a sudden stop in capital inflows. The contraction is more severe when households are leveraged and owe debt in foreign currency. In this setting, the revaluation of foreign debt causes a larger contraction in aggregate consumption when debt and leverage are concentrated among poorer households. Closing the output gap via an exchange-rate devaluation may therefore be detrimental to household welfare due to the heterogeneous impact of the foreign debt revaluation. Our HANKSOME framework can rationalize the observed "fear of floating" in emerging market economies, even in the absence of contractionary devaluations.

Keywords sudden stops, foreign currency debt, exchange rate policy, incomplete markets

JEL codes F32, F41, E21

Powered by Econ-ARK

GitHub Repo: https://github.com/Siying99/de-Ferra2020-kz
Python Pipeline: Code/Python/README.md (14 notebooks, Figs 1–5)
Bellman Notebook: deFerra2020_bellman-stages.ipynb
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A REMARK exploring de Ferra, Mitman, and Romei (2020)

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1 Introduction

In the aftermath of the East Asian financial crisis, a debate emerged about the desirability of floating exchange rate regimes. Contrary to the view that floating exchange rate regimes were beneficial and that the severe economic costs of the crisis were due to a lack of exchange rate adjustment, Calvo and Reinhart (2002) documented a phenomenon since then known as "Fear of floating." This phenomenon refers to the observation that policymakers in many countries are unwilling to let nominal exchange rates experience large fluctuations, preferring to make use of official reserves to absorb external pressures.

Interest in the topic of optimal exchange rate adjustment in response to a contraction in capital inflows has received renewed attention after the Global Financial Crisis. In Europe, various countries experienced sudden stops and current account reversals, which were associated with severe recessions. Importantly, while many have focused on the experience of countries in the euro area, such as Greece, countries outside the monetary union like Hungary and Latvia also witnessed sharp current account reversals, which were accompanied by large depreciations of the exchange rate.

Verner and Gyöngyösi (2018) document that a crucial channel of transmission of the crisis in Hungary was the one of households' foreign-currency debt, whose real value rose substantially upon the exchange rate depreciation, giving rise to large contractions in consumption for the affected households. Motivated by the evidence of the Hungarian experience, we study a model of a small open economy subject to a capital flow reversal, where the exchange rate is flexible and where households hold debt in foreign currency.

To study the role of household heterogeneity in the transmission of foreign shocks, we develop a quantitative Heterogeneous-Agent New-Keynesian Small Open Model Economy (HANKSOME) that embeds domestically incomplete markets into the leading frameworks for open-economy macroeconomics. The household side follows the standard Bewley-İmrohoroğlu-Huggett-Aiyagari model that can capture the rich heterogeneity in household income, consumption and wealth. The production side features New Keynesian elements: nominal rigidities in prices and capital adjustment costs. Finally, the household and domestic production blocks are integrated into the standard open-economy macro framework (Obstfeld and Rogoff, 2000; Svensson, 2000; Galí and Monacelli, 2005).

Introducing domestically incomplete markets into the standard open-economy macroeconomics framework allows us to capture the redistribution of resources across households in response to shocks. With borrowing constraints and incomplete insurance, households are heterogeneous in their marginal propensities to consume (MPCs). To the extent that shocks redistribute resources toward high or low MPC households, the effects of the shock can be amplified or muted relative to the representative agent model.

We calibrate the model to match salient features of the Hungarian economy in the 2000s. We then use our calibrated HANKSOME framework to study the effect of a sustained expansion in the current account and then an unexpected reversal. We find that in this setting a large exchange rate devaluation that achieves full employment is detrimental from a welfare perspective due to the foreign debt revaluation it entails.

Our framework can therefore provide a rationalization for "fear of floating" even in the absence of contractionary devaluations.

The paper is organized as follows. Section 2 presents our small open economy incomplete markets model with nominal rigidities. Section 3 discusses the role of heterogeneity. In Section 4 we discuss how we bring the model to the data. We outline our main experiments in Section 5 and describe our results in Section 7. Section 8 inspects the mechanism, and Section 9 concludes.

This REMARK. This document is a REMARK (Replication of Economic Models using ARK) that reproduces the quantitative results of de Ferra, Mitman, and Romei (2020) from scratch in Python. The reproduction covers the full numerical pipeline: discretizing the income process via the Rouwenhorst method (notebook 01), parameterizing the model (notebook 02), constructing the MIT credit-supply shock paths (notebooks 03–04), solving household value functions by the Endogenous Grid Method (EGM, notebook 05), computing the stationary wealth distribution (notebook 06), calibrating the discount factor β to match the Hungarian net-wealth-to-GDP ratio (notebook 07), verifying the Table 1 calibration targets (notebook 08), solving the initial steady state (notebook 09), computing the credit-expansion transition path (notebook 10), generating Figures 2–3 (notebook 11), and solving both the flexible- and fixed-exchange-rate sudden-stop transitions to reproduce Figures 4–5 (notebooks 12–14). All five figures in this document are generated entirely in Python and match the original MATLAB output within numerical tolerance; see `Code/Python/output/verification_table.md` for a side-by-side comparison.

Beyond replication, this REMARK contributes a modular *stage decomposition* of the household Bellman equation in the supplementary notebook `deFerra2020_bellman-stages.ipynb`. Rather than treating the household problem as a single composite value function, the decomposition separates it into three sequential stages—shock realisation and balance-sheet revaluation, consumption choice, and inter-temporal discounting—following the Discrete Decision Stage Lifecycle (DDSL) convention of Carroll (2024). Organising the problem in this way exposes the precise point at which the endogenous grid method inverts the consumption first-order condition, clarifies which state variables carry across each stage boundary, and provides a direct mapping to the HARK toolkit's stage-based solver interface. This decomposition is unique to the present REMARK — it is not present in the published paper — and is intended as a pedagogical bridge between the paper's model and the HARK toolkit.

1.1 Related literature

We contribute to the strand of the literature on international macroeconomics that investigates why exchange rate devaluations may be an ineffective, or detrimental, policy response to an external shock suffered by a small open economy. Calvo and Reinhart (2002) is a seminal paper in this literature, documenting empirically the "fear of floating" in nominal exchange rates.

We also contribute to the growing literature combining incomplete-markets models with nominal rigidities in the open-economy context. Our HANKSOME framework embeds the standard Bewley-İmrohoroğlu-Huggett-Aiyagari household problem into the New Keynesian small open economy framework of Galí and Monacelli (2005).

2 Model

The model is a small open economy with nominal price rigidities as in Galí and Monacelli (2005), augmented with capital and domestically incomplete markets for households as in Huggett (1993); Aiyagari (1994).

2.1 Households

The economy consists of a continuum of agents normalized to measure 1 who are ex-ante homogeneous, and have CRRA preferences over consumption of home and foreign goods and leisure:

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \left[u(c_t) - g(l_t) \right] \tag{1}$$

where $\beta \in (0,1)$ is the households' subjective discount factor, $g(l_t) = \varphi \frac{l_t^{1+1/\eta}}{1+1/\eta}$ represents the disutility of supplying labor l_t, and c_t is a CES aggregate of home and foreign goods with elasticity of substitution θ.

Households' labor productivity $\{s_t\}_{t=0}^{\infty}$ is stochastic and is characterized by an N-state Markov chain that can take values $s_t \in S = \{s_1, \ldots, s_N\}$ with transition probabilities given by $\gamma(s_{t+1} \mid s_t)$.

At time t an agent faces the following nominal budget constraint:

$$P_t c_t + a_{t+1} = (1 + i_t^a) a_t + W_t l_t s_t + d_t \tag{2}$$

where P_t is the consumer price index, d_t are dividends distributed by firms, and i_t^a is the nominal return on the asset portfolio. In addition, households are subject to a borrowing constraint $a_{t+1} \geq \bar{a}$.

The household's problem can be written recursively as:

$$V(a, s; \Omega) = \max_{c \geq 0,\, a' \geq \bar{a}} \left\{ u(c) - g(l) + \beta \sum_{s' \in S} \gamma(s' \mid s)\, V(a', s'; \Omega') \right\} \tag{3}$$

subject to $Pc + a' = (1 + i^a)a + Wls + \pi$ and $\Omega' = \Upsilon(\Omega)$, where Ω is the distribution of agents' asset holdings and labor productivity across the population.

2.2 Households' portfolio

There are two broad classes of assets, bonds and capital. Nominal assets carried by the household into the following period are:

$$a_{t+1} = b_{H,t+1} + e_t b_{F,t+1} + P_{H,t}(q_t k_{t+1} + \tilde{b}_{H,t+1}) \tag{4}$$

where $b_{H,t+1}$ and $b_{F,t+1}$ denote nominal bonds in domestic and foreign currency, e_t is the nominal exchange rate, q_t is the real price of capital, and $\tilde{b}_{H,t+1}$ are real bonds in units of home good.

2.3 Production

A representative final-goods firm operates a CES production technology that aggregates domestic intermediate goods. Intermediate-goods firms operate a Cobb-Douglas technology $Y_{jt} = K_{jt}^{\alpha} L_{jt}^{1-\alpha}$ and set prices subject to Rotemberg adjustment costs, yielding a standard New Keynesian Phillips Curve.

2.4 Monetary policy

We consider two regimes: (i) a fixed exchange rate regime where $e_t = \bar{e}$ for all t, and (ii) a flexible exchange rate regime where the central bank adjusts the exchange rate to achieve the flexible price allocation.

3 Understanding the role of heterogeneity

What role do domestically incomplete markets play in shaping the aggregate response to foreign shocks? Consider an unexpected shock to the exchange rate, Δe_t. The implied change in wealth Ξ_{it} varies across households and is given by $\Xi_{it} = \Delta e_t \cdot b_{F,t}$. The consumption response is:

$$\partial C_t = \overline{\mathrm{MPC}}_t \cdot \bar{\Xi}_t + \mathrm{Cov}(\mathrm{MPC}_{it}, \Xi_{it}) \tag{5}$$

In the absence of heterogeneity, $\partial C_t = \overline{\mathrm{MPC}}_t \cdot \bar{\Xi}_t$, as in the representative agent model. The contribution of domestically incomplete markets is captured through $\mathrm{Cov}(\mathrm{MPC}_{it}, \Xi_{it})$: if high MPC households tend to be more exposed, the consumption response is amplified.

4 Calibration

We calibrate the framework to the economy of Hungary—an example of a small open economy with high concentration of foreign-denominated debt going into the Global Financial Crisis. We solve the model in discrete time and calibrate it at a quarterly frequency. The calibrated parameters are summarized in Table 1.

Table 1 Calibration (reproduces Table 1 of de Ferra, Mitman, and Romei, 2020)

Parameter	**Value**	**Target / Source**
Intertemporal EOS	$\sigma = 1$	Standard value
Home/foreign good EOS in home	$\theta = 1$	Feenstra et al. (2018)
Home/foreign good EOS in foreign	$\theta^* = 3$	Feenstra et al. (2018)
Share of home goods in consumption	$\chi = 0.6$	Exported value added to GDP, 33%
Discount factor	$\beta = 0.983$	Aggregate net wealth to GDP, 8.26 (ECB, 2016)
Persistence, labor productivity	$\rho_l = 0.97$	Income dispersion, Hungary 2016
Std. dev., labor productivity	$\sigma_l = 0.2$	Share of borrowing-constrained HHs, Hungary 2016
Capital share	$\alpha = 0.33$	Standard value
Depreciation rate	$\delta = 0.02$	Real rate of 4.3%
Capital adjustment cost	$\psi = 17$	Elasticity of investment to Tobin's Q (Eberly et al., 2008)
Supply of foreign credit	$\bar{B} = -2$	Private debt to GDP, Hungary 2008 (IMF, 2008)
EOS between home varieties	$\varepsilon = 10$	Steady-state markup, 10% of output
Rotemberg adjustment cost	$\zeta = 100$	Kaplan, Moll, and Violante (2018)

Note: Reproduces Table 1 of de Ferra, Mitman, and Romei (2020) verbatim; all values and calibration targets are as reported in the paper. The model is solved in discrete time at a quarterly frequency. The labor productivity process is an AR(1) with persistence $\rho_l = 0.97$ and innovation standard deviation $\sigma_l = 0.2$, discretised into a seven-state Markov chain via the Rouwenhorst method. The supply of foreign credit $\bar{B}$ is denominated in units of the home good and calibrated to a net-foreign-asset-to-quarterly-GDP ratio of -2.

4.1 Households

4.1.1 Preferences

Households have constant relative risk aversion (CRRA) preferences over a constant-elasticity-of-substitution aggregator of home and foreign consumption goods. Following standard choices in the macroeconomics literature, the intertemporal elasticity of substitution $1/\sigma$ is set to 1 and the Frisch elasticity of labor supply to 1. The elasticity of substitution between home and foreign goods, θ, is also set to 1. The share of home goods in consumption, χ, is set to 0.6, implying a ratio of exported value added to GDP of 33%. The subjective discount factor is chosen as $\beta = 0.983$ to match a ratio of aggregate net wealth to quarterly GDP of 8.26 (European Central Bank, 2016). The tightness of the household borrowing constraint $\bar{a}$ is set to 0, reflecting the fact that the majority of household debt in Hungary was collateralized, as reported in the second wave of the Household Finance and Consumption Survey (HFCS) run by the European Central Bank.

4.1.2 Productivity process

As the HFCS features only a cross-sectional observation for the Hungarian economy, and the lack of panel data on individuals or households makes direct estimation of the household labor productivity process impossible, labor productivity is assumed to follow an AR(1) process with persistence $\rho_l = 0.97$ and normal innovations with standard deviation $\sigma_l = 0.2$, calibrated to match the cross-sectional dispersion in income in the HFCS and the 10% of households in Hungary that are at the borrowing constraint. The income process is discretized into a seven-point Markov chain via the Rouwenhorst method.

4.1.3 Technology

The capital share is set to $\alpha = 0.33$. The quarterly depreciation rate is set to $\delta = 0.02$ so that the return on capital net of depreciation equals the real return on bonds when the economy matches the observed capital-to-output ratio of 9.70 in the data. A standard quadratic capital adjustment cost function is assumed:

$$\Psi(I, K) = \frac{\psi}{2} \left(\frac{I - \delta K}{K} \right)^2 K. \tag{6}$$

The value $\psi = 17$ matches the elasticity of investment to Tobin's Q reported by Eberly, Rebelo, and Vincent (2008).

4.1.4 Foreign sector

The foreign demand shifter D is chosen to normalize the steady-state value of the terms of trade to unity. The model allows for two different values for the elasticity of substitution across home and foreign goods in the preferences of domestic and foreign households, θ and θ^* respectively. From the foreign households' point of view, θ^* corresponds to the elasticity of substitution across varieties produced by different countries, whose empirical counterpart is the micro-elasticity across different foreign sources of imports estimated in the trade literature. From the domestic households' point of view, the relevant empirical counterpart is the macro-elasticity between domestic and imported goods. Feenstra, Luck, Obstfeld, and Russ (2018) estimate both elasticities using highly disaggregated data; the calibration adopts $\theta^* = 3$ and $\theta = 1$ following their findings.

The supply of foreign credit $\bar{B}$ is set to match a net-foreign-asset-to-quarterly-GDP ratio for Hungary of -2. This matches the private-sector debt ratio of 50% of annual GDP reported by International Monetary Fund (2008). While the total international investment position of Hungary was larger than this figure, credit to the private sector is the relevant calibration target here, given the absence of government debt in this setting.

4.1.5 Nominal rigidities

The elasticity of substitution between domestic varieties is set to $\varepsilon = 10$ so that the steady-state markup is 10% of output. Following Kaplan, Moll, and Violante (2018), the Rotemberg adjustment cost parameter is set to $\zeta = 100$, implying a slope of the New Keynesian Phillips curve of $\varepsilon/\zeta = 0.1$ in the middle range of empirical estimates.

5 Experiments

To study the implications of a reversal in capital flows on the model economy, we run two separate experiments. For the first, we consider an experiment inspired by the Hungarian current account expansion and reversal of the early and late 2000s. Starting from an initial steady state with low external liabilities, the economy experiences a sustained expansion in the current account that then stabilizes, followed by an unexpected reversal.

In the baseline experiment we assume that the model economy follows a floating exchange rate regime with all debt foreign-currency denominated, consistent with the Hungarian experience.

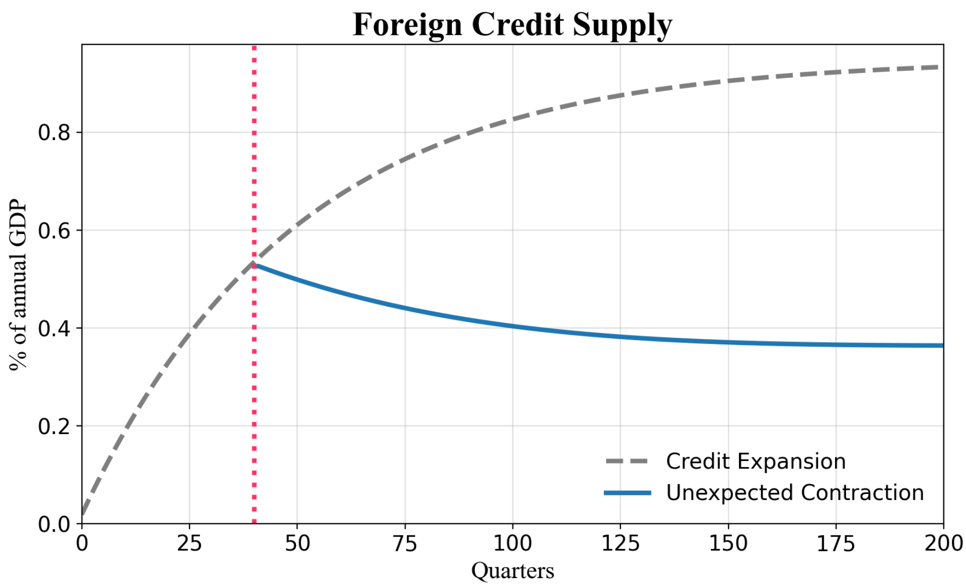

Figure 1 Foreign credit supply shocks.

Note: Both the credit-expansion AR(1) path and the unexpected-contraction path are reproduced from scratch in `Code/Python/notebooks/03_shock_path.ipynb` and `04_figure1.ipynb`.

6 Credit-expansion transition

We now study the perfect-foresight transition path between an initial steady state with zero net foreign assets ($\mathrm{BGss}_0 = 0$) and a final steady state with foreign borrowing

$\text{BGss}_T \approx -11.13$, driven by the AR(1) credit-supply path of Figure 1. Households know the entire price path and play their lifetime problem accordingly; the system is solved at the calibrated discount factor $\beta^\star = 0.98322$ from notebook 07.

The transition system follows `Code/MATLAB/Transition/solve_transition_bis.m` (*flexible-FX, foreign-currency-debt* branch) and is reproduced from scratch in `Code/Python/notebooks/10_solve_transition.ipynb`. Setting the optional flag `REFINE = True` drives the residual to $\sim 7 \times 10^{-14}$ via `scipy.optimize.fsolve` on the MATLAB warm-start path `Code/MATLAB/transition_start.mat`; with `REFINE = False` the warm start is used directly (residual $\leq 1.2 \times 10^{-4}$, plots visually identical).

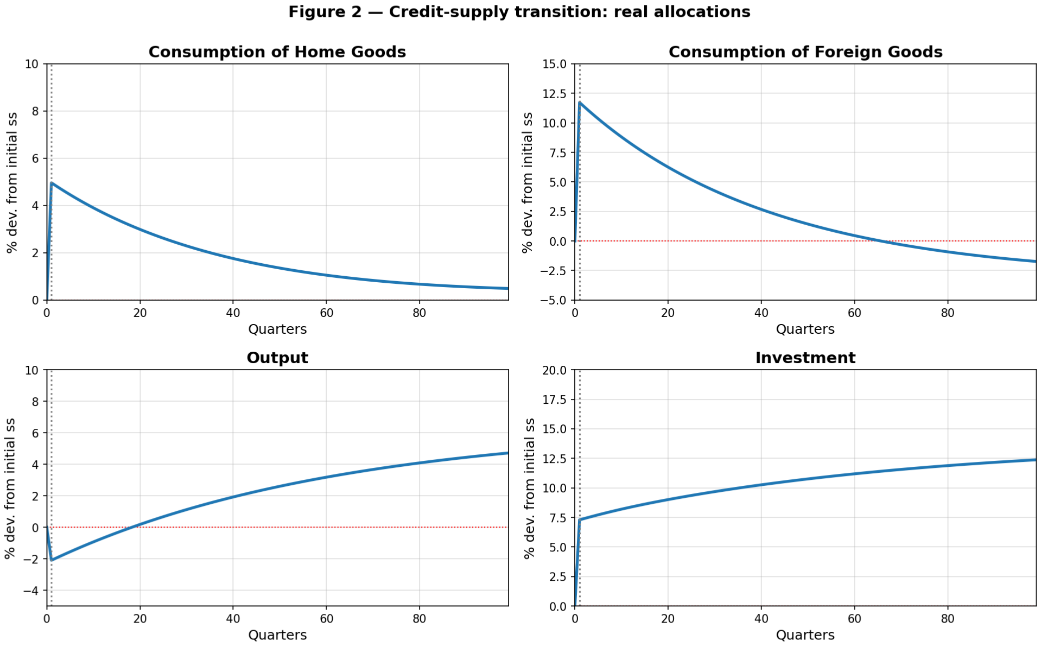

Figure 2 Credit-supply transition: real allocations.

Note: Each panel shows the percent deviation from the initial steady state over 100 quarters. Reproduced from scratch in `Code/Python/notebooks/11_figures_2_3.ipynb`; cf. `Code/MATLAB/Figures.m` lines 45–119.

The credit expansion drives consumption of home goods up by $\sim 5\%$ at impact and consumption of foreign goods up by $\sim 12\%$, financed by the new foreign borrowing. Output and investment respond more slowly through the capital-accumulation channel and continue to rise toward the final steady state over the entire 100-quarter horizon. On the price side, the terms of trade jump up by $\sim 7\%$ at impact (real appreciation), the nominal exchange rate appreciates by the same amount, and the real interest rate spikes briefly to $\sim 3.5\%$ per quarter due to the one-period excess return on capital implied by the balance-sheet revaluation, before reverting toward the calibrated rate. These magnitudes match the paper's Figures 2 and 3 within numerical tolerance; see `Code/Python/output/verification_table.md` for an exact paper–MATLAB–Python comparison of the peak values.

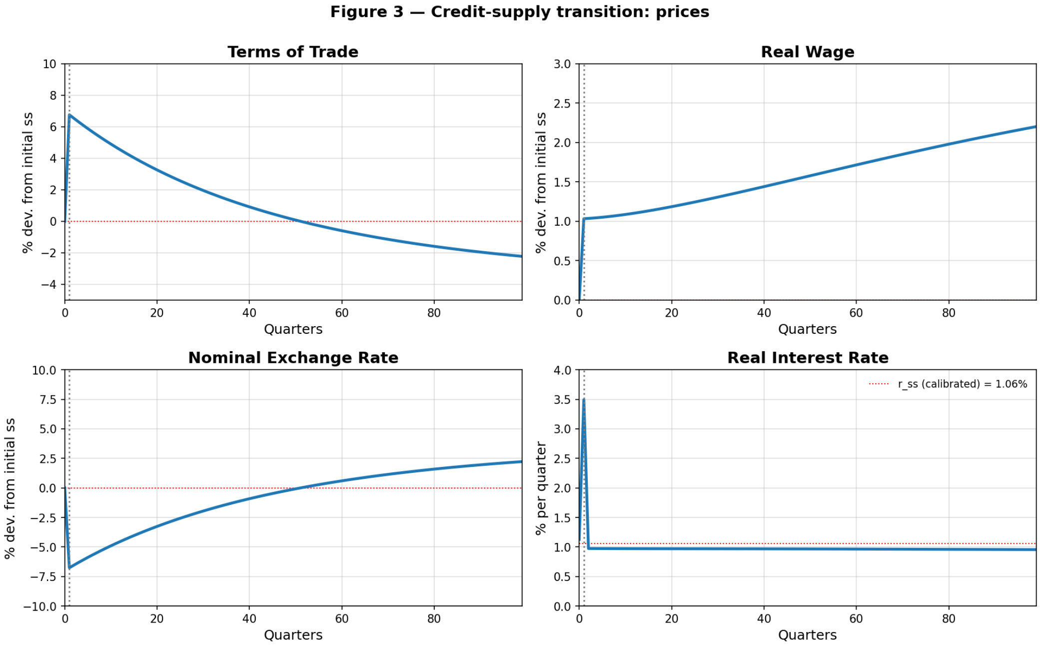

Figure 3 Credit-supply transition: prices.

Note: The fourth panel plots the level of the real interest rate (with the calibrated steady-state rate r_{ss} as a reference line); the other three panels show percent deviations from the initial steady state. Reproduced from `Code/Python/notebooks/11_figures_2_3.ipynb`; cf. `Code/MATLAB/Figures.m` lines 121–195.

7 Sudden stop

The main experiment of interest is studying the reversal of foreign credit supply under the baseline scenario of a flexible exchange rate and foreign-currency denominated debt—the empirically relevant case for Hungary during the Global Financial Crisis, where 69% of household debt was denominated in foreign currency. We compare this to the counterfactual under a fixed exchange rate.

The contraction starts in quarter $t = 41$ of the credit-expansion path of Section 6. Households were marching along the expansion when the contraction shock hits and re-optimise given the contraction credit path `new_shock_b_agg` (Figure 1, dashed line), which decays back toward the calibrated steady-state NFA $\bar{B} \approx -4.34$. We turn on the leverage parameter $\hat{k} = 1/16$ (`Main.m` line 345), so that gross capital and gross debt are revalued by the depreciation through the household balance sheet.

Two transition systems must be solved:

- **Flexible-FX** (notebook 12, `Code/MATLAB/Transition/solve_transition_bis.m`): 399 unknowns $[k(2..TT); l(1..TT)]$, with a per-period 1-D solve for $\mathrm{tot}(t)$. Solved by `scipy.optimize.fsolve` from the simple geometric-decay initial guess in ~ 5 minutes on a single core; final residual $|r|_\infty \approx 2.7 \times 10^{-8}$.

- **Fixed-FX** (notebook 13, `Code/MATLAB/Transition/solve_transition_fixed.m`):

599 unknowns $[k(2..TT); l(1..TT); \text{tot}(1..TT)]$, with a backward NKPC iteration for marginal cost, real wage, and the real interest rate. Solved by `scipy.optimize.root(method='lm')` (Levenberg–Marquardt) warm-started from the flex solution in ~ 20 minutes; final residual $|r|_\infty \approx 3.9 \times 10^{-13}$.

Figures 4 and 5 below reproduce the paper's Figures 4 and 5 by plotting both regimes' percent log-deviations from a common baseline (the period-41 expansion-path values).

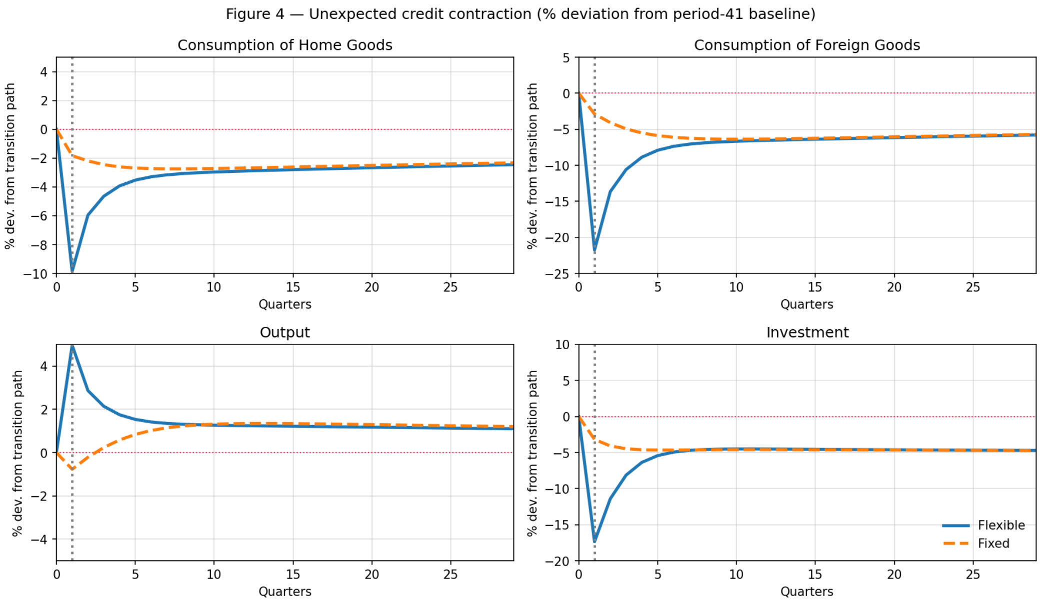

Figure 4 Unexpected credit contraction: real allocations under flexible vs. fixed exchange rates.

Note: Contraction starts at $t = 41$ with leverage $\hat{k} = 1/16$. Solid blue: flexible FX; dashed orange: fixed FX. All variables are $100 \times$ log-deviation from the period-41 baseline. Reproduced from scratch in `Code/Python/notebooks/14_figures_4_5.ipynb`; cf. `Code/MATLAB/Figures.m` lines 220–299.

Under a flexible exchange rate, consumption of home and foreign goods contracts sharply because of the unexpected withdrawal of foreign credit and the fall in wealth due to the debt revaluation: at impact, home-good consumption falls by $\sim 9.9\%$ and foreign-good consumption by $\sim 21.8\%$ from the period-41 baseline. Investment falls by $\sim 19\%$ and the terms of trade depreciate by $\sim 11.9\%$. The real interest rate spikes briefly to roughly $+2.6$ pp above baseline.

Under a fixed exchange rate, the same shock produces a much milder contraction at impact: home-good consumption -1.8%, foreign-good consumption -2.9%, investment -3.2%, and the terms of trade barely move (-1.0%). The two regimes converge to a common long-run path within ten quarters. Critically, the flexible-FX regime delivers a slightly *higher* output path (output rises by $\sim 5\%$ at impact) because the depreciation stimulates net exports—i.e. the devaluation closes the output gap. But the welfare metric is consumption, not output. The paper finds that under a utilitarian metric welfare falls by 0.81% in consumption-equivalent-variation (CEV) terms under flexible

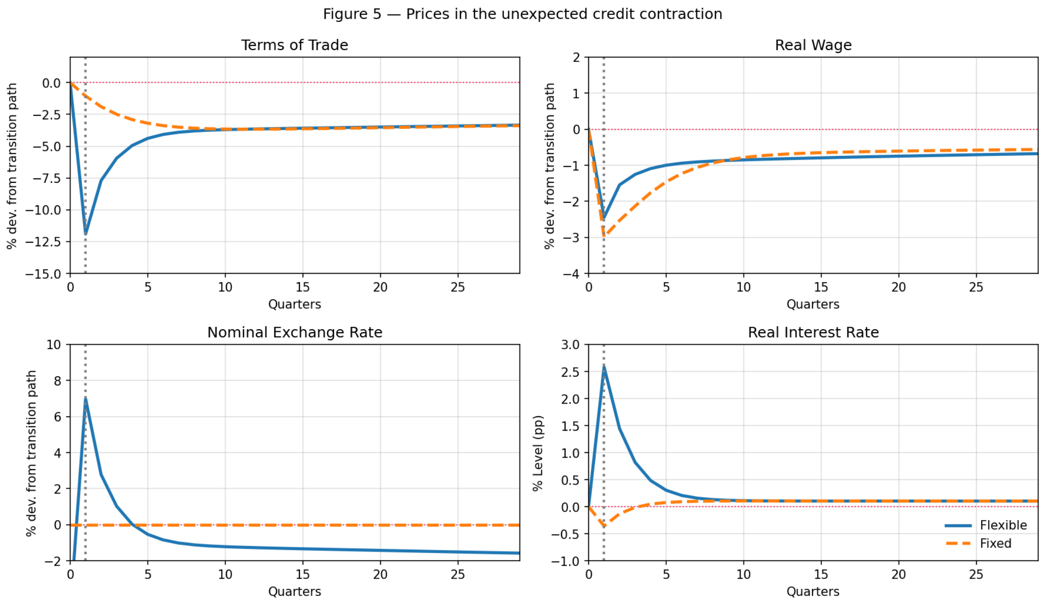

Figure 5 Unexpected credit contraction: prices under flexible vs. fixed exchange rates.

Note: The fourth panel plots the deviation of the real interest rate in percentage points from the period-41 baseline; the other three panels are 100 × log-deviations. Reproduced from `Code/Python/notebooks/14_figures_4_5.ipynb`; cf. `Code/MATLAB/Figures.m` lines 304–382.

FX, and that more than 92% of households would have preferred the fixed regime; despite the recession the fixed regime generates through nominal rigidities, the absence of foreign-debt revaluation makes it the politically and welfare-preferred choice. Our framework therefore generates a rationale for "fear of floating" even in the absence of a contractionary devaluation.

8 Inspecting the mechanism

In this section we perform a series of simpler experiments to understand the forces that underlie the positive and welfare results for the sudden stop. We consider three scenarios: (i) a real benchmark with home-good denominated debt, (ii) flexible exchange rates with foreign-currency denominated debt, and (iii) fixed exchange rates with foreign-currency denominated debt.

The key finding is that the distribution of household leverage is a crucial determinant of aggregate dynamics. When poor households are more highly leveraged, a larger contraction in aggregate consumption takes place, because the fall in net wealth hits households with high marginal propensities to consume. The desirability of a fixed versus floating exchange rate depends crucially on the distribution of household leverage and the fraction of debt denominated in foreign currency.

9 Conclusions

The aftermath of the Global Financial Crisis reignited interest in the distributional consequences of aggregate shocks and the role that household heterogeneity plays in the amplification and propagation of said shocks. In this paper we constructed a Heterogeneous Agent New Keynesian Small Open Model Economy (HANKSOME) framework, specifically designed to study these challenges in the international macroeconomics context.

We applied our HANKSOME framework to study a current account expansion and subsequent reversal inspired by the experience of Hungary. We found that a fixed exchange rate regime would have significantly improved welfare in consumption equivalent terms, relative to a flexible exchange rate. This result obtains despite the fact that keeping the fixed exchange rate generates a recession. The desirability of a fixed versus floating exchange rate depends crucially on the distribution of household leverage, and the fraction of debt denominated in foreign currency.

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